

Part 1 Rocks – part of the Earth system

The core activities in this unit total about 22 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	Every rock tells a story	ACSSU153 ACSHE 134 AC SIS148	Engage & Explore	In an animation Connie Conglomerate introduces some of the rocky characters, and they each tell a little about their history.	Core
			Digital		
			Short		
1.2	The deep Earth	ACSSU153 ACSHE 134 ACSHE136 AC SIS148	Engage & Explore	Introduces the inner structure of the Earth.	Core
			Digital		
			Short		
1.3	Cycling rocks	ACSSU153 AC SIS145 AC SIS148	Explore & Explain	A first introduction to the idea that rocks change and are recycled through Earth's processes.	Core
			Digital		
			Short		
1.4	It's all in the crust	ACSSU153 AC SIS144 AC SIS145 AC SIS148	Explore & Explain	Students examine and describe basic rock types (sedimentary, igneous and metamorphic) according to their characteristics and relate them to their processes of formation.	Core
			Hands-on & Digital		
			Medium		

Part 2 Minerals – ingredients in the rock recipe

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Rock or mineral?	ACSSU153 ACSIS145 ACSIS148	Explore & Explain	Students look closely at some rocks to discover their mineral composition and determine the difference between rocks and minerals.	Core
			Hands-on & Digital		
			Short		
2.2	Diagnostic properties	ACSSU153 ACSIS145 ACSIS148	Explore & Explain	Students examine various minerals and determine diagnostic properties of minerals that can be used to tell minerals apart.	Optional
			Hands-on & Digital		
			Medium		
2.3	Testing minerals	ACSSU153 ACSIS145 ACSIS234 ACSIS148	Engage, Explore, Explain & Elaborate	Students are introduced to mineral tests and use them to differentiate between biotite and quartz. Students then use the tests to identify two unknown mineral samples.	Core
			Hands-on		
			Medium		
2.4	Rock identification	ACSSU153 ACSIS145 ACSIS148	Explain & Elaborate	Students identify the minerals in three rock specimens and then use this information to identify the three rocks.	Core
			Digital		
			Short		
2.5	Mining our everyday minerals	ACSSU153 ACSHE 134 ACSHE 135 ACSHE136 ACSIS148	Explore	Students learn how important minerals are in producing many of our everyday objects and learn to associate various minerals with their uses. Students study Australia's export industries and appreciate the large role minerals and fossil fuels make to our economy.	Core
			Digital		
			Short		

Part 2 Minerals – ingredients in the rock recipe (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.6	Careers in mineral exploration	ACSSU153 ACSHE 134 ACSHE 135 ACSHE136 ACSIS148	Explore & Explain	Students explore the challenges of extracting valuable ores and minerals from their parent rocks. Students explore the various career pathways in the mining industry and learn where deposits and mines are located in Australia.	Core
			Digital		
			Long		
2.7	Money for rocks	ACSHE136 ACSIS148	Explain & Elaborate	Students are presented with five minerals and asked which contributes most to Australia's economy.	Optional
			Digital		
			Short		

Part 3 Rocks from the Earth's Furnace – igneous rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	Big and small crystals	ACSSU153 ACSIS140 ACSIS145 ACSIS234 ACSIS148	Engage, Explore, Explain & Elaborate	Students investigate the effect of temperature on the formation of salol crystals and relate their experience to the formation of igneous rocks.	Core
			Hands-On		
			Medium		
3.2	Erupting and intruding	ACSSU153 ACSIS148	Explore, Explain & Evaluate	Students learn about the distinction between volcanic and plutonic rocks. They also learn the terms 'extrusive' and 'intrusive'.	Core
			Hands-on & Digital		
			Medium		

Part 4 Grain by grain – sedimentary rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Breaking rocks	ACSSU153 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain	Students simulate the processes of physical and chemical weathering using sugar cubes. They further explore weathering and erosion in a video.	Core
			Hands-on & Digital		
			Long		
4.2	Which sediments where?	ACSSU153 ACSIS145 ACSIS148	Explore, Explain & Elaborate	Students explore different sedimentary environments and relate this to their own region.	Core
			Digital		
			Short		
4.3	Sticking things together	ACSSU153 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain	Students make various rocks from a range of sediments and binders. Students design fair tests to determine which rocks are harder and more durable. Students record their results in both a table and a graph and draw conclusions from their results.	Core
			Hands-on		
			Long		
4.4	Grains and rocks	ACSSU153 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore & Explain	Students examine three different sedimentary rocks and identify their key characteristics.	Core
			Hands-on & Digital		
			Medium		
4.5	Making rocks from body parts	ACSSU153 ACSHE 135 ACSIS141 ACSIS144	Explore & Explain	Students explore the formation and characteristics of some biogenic rocks, including limestone and coal.	Core
			Hands-on & Digital		
			Medium		

Part 5 Baking and burying: metamorphic rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	These rocks are so hot	ACSSU153 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain	Students bake modelling clay and compare its characteristics with unbaked clay. They relate this to the formation of some metamorphic rocks.	Core
			Hands-on & Digital		
			Long		
5.2	Regional metamorphosis	ACSSU153 ACSIS145 ACSIS234 ACSIS148	Explore & Explain	Students read about the famous tilted beds of metamorphic rocks in northern Scotland that extend for about 250 km and see the scale on which burial can occur; leading to large scale regional metamorphism. Students conduct a simple acid fizz test to identify that limestone and marble have similar chemical properties.	Core
			Hands-on & Digital		
			Short		
5.3	A tortured history	ACSSU153 ACSHE136 ACSIS148	Explain & Elaborate	This activity summarises the ideas of contact metamorphism and regional metamorphism into a simple diagram that illustrates which rocks form due to increased temperature or pressure, or both. Students learn about the life of Neville Gneiss and watch two more short videos that explore metamorphic rocks.	Core
			Digital		
			Short		

Part 6 The rock cycle and geological time

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.1	Rocks into rocks	ACSSU153 ACSHE226 ACSHE136 ACSIS146 ACSIS148	Engage, Explore, Explain, Elaborate & Evaluate	Students consolidate their understanding through activities exploring the rock cycle. These include an on-line interactive quiz and a hands-on activity modelling different aspects of the rock cycle.	Core
			Hands-on & Digital		
			Long		
6.2	On the level	ACSSU153 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore & Explain	This is a hands-on activity where students simulate the deposition of sediments to investigate: - the relative ages of sediments, - the horizontal orientation of the beds and - the nature of graded bedding.	Optional
			Hands-on		
			Medium		
6.3	How do fossils form?	ACSSU153 ACSHE 134 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain	Students complete an interactive exercise illustrating the formation of fossils. In the Notebook exercise students learn about South Australia's world heritage listed Naracoorte Caves and the fossils found there of Australia's ancient megafauna. Students watch a video (6 minutes) on palaeontology in Australia which introduces stratigraphy (which is further developed in Activity 6.4). In the Find out more section students may explore the range of fossils found in Australia by exploring links to the Queensland and Melbourne museums.	Core
			Digital		
			Short		
6.4	Is this the same as that?	ACSSU153 ACSIS146 ACSIS234 ACSIS148	Explore, Explain & Elaborate	Students work in small groups to sequence cards that represent different rock layers containing fossils. Sequencing the rock layers simulates how palaeontologists use fossils to achieve stratigraphic correlation.	Core
			Classroom		
			Medium		

Part 6 The rock cycle and geological time (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.5	Relative geological time	ACSSU153 ACSIS148	Explore, Explain & Elaborate	Students explore the horizontal bedding and sheer majesty of the Grand Canyon. Using age data for each of the layers, they map onto a diagram of the Grand Canyon the name of the geological period when the layer was deposited and possible fossils that could be found from that period.	Core
			Digital & Classroom		
			Medium		
6.6	How long does it take?	ACSSU153 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore, Explain & Elaborate	Students roll dice in the Rock Cycle Game to follow their rock through the rock cycle and experience the magnitude of geological time. This activity links students' understanding of the Rock Cycle with their understanding of the Relative Geological Time Scale.	Core
			Digital & Classroom		
			Medium		
6.7	Forming landforms	ACSSU153	Explain & Elaborate	Students investigate several Australian landforms and the different environmental processes that led to their formation.	Core
			Digital		
			Short		
6.8	Sample test		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Long		

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